

Who gains from AI?

Autonomy. Capability. Proof.

Ide & Talamàs (2025)

github.com/gsaco/ai-05-ide

Read: course PDF, May 20, 2025 (39 pp). Lean pin: arXiv v11, February 25, 2025 (35 pp).

Two dimensions. Three outcomes.

AUTONOMY = PERMITTED ROLES

Autonomous AI

Produces, works, and advises.

Co-pilot-only AI

Advises; cannot initiate production.

CAPABILITY = KNOWLEDGE

$$a = z_{AI} \in [0, 1).$$

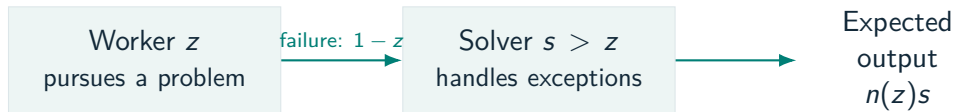
The fraction of problems AI can solve.

A permitted role may still be unprofitable.

Track **bottom wages**, **top wages**, and **total output**. The sign cannot be inferred from autonomy alone.

Today: source claims → a three-type economy → seven slides on what Lean checks.

Knowledge helps only when someone supplies time



$$h n(z)(1 - z) = 1 \implies n(z) = \frac{1}{h(1 - z)}.$$

$$\max_{\text{organization, } z, s} \Pi, \quad \Pi_{HH} = n(z)[s - w(z)] - w(s).$$

Humans maximize income; firms choose teams; competitive entry sets active profits to zero.
Labor and compute markets clear.

Section 3.1. Difficulty is uniform on $[0, 1]$; each request costs h solver time, successful or not.

Keep the conditions attached to the result

PEOPLE AND FIRMS

Unit human mass; one unit of time each.
Observable knowledge $z \in [0, 1]$ with
continuous, strictly positive density.

Independent uniform difficulties; risk neutrality;
output price one. Competition, complete
information, no fixed costs; at most two layers.

The source's top-winner result needs **both** $h < h_0$ **and** $a < 1$. The appendix gives the sufficient compute bound.

Sections 3.1, 3.3, 5.2 and 6. These continuum assumptions are not inherited by the finite example.

TECHNOLOGY AND SCARCITY

$0 < h < h_0$: no independent humans before
AI.

$a < 1$: perfect AI is excluded.

Compute abundant relative to human time;
opportunities exceed production capacity.

Compare autonomy at the same a and μ .

Proposition 5: capability creates bottom winners

BOTTOM: A STRICT THRESHOLD

$$B \neq \emptyset \iff a > \bar{a}, \quad \bar{a} \in \text{int } W.$$

W : the pre-AI worker set. B : humans below AI who earn more than before.

The threshold lies *inside* the worker region: some “basic” AI already benefits the bottom.

TOP: ALWAYS SOME WINNERS

$$T \neq \emptyset \quad \forall a \in [0, 1).$$

Under the stated conditions, some humans above AI gain.

The human with knowledge a loses.

Autonomy is held fixed in this proposition. **Capability still changes the distributional conclusion.**

$B = \{z \leq a : w^A(z) > w(z)\}$, $T = \{z \geq a : w^A(z) > w(z)\}$ within $[0, 1]$. v11 p.24 / course p.25.

Proposition 6: free advice must still be useful

Capability	Co-pilot-only equilibrium, $r^N = 0$
$a \leq w(0)$	No adoption. Wages and occupations unchanged.
$a > w(0)$	Lowest-knowledge humans adopt. Strict bottom gains and some strict human losses.

NEAR THE BOTTOM

$w^N \geq \max\{w, w^A\}$; strict if $a > w(0)$.

NEAR THE TOP

$w^N \leq w^A$; strict away from $z = 1$.

TOTAL OUTPUT

$Y^A > Y^N$. Efficiency is relative to the permitted technology.

$w(0)$ is an **adoption threshold**; $\bar{a}$ is an **autonomous bottom-gain threshold**. They are distinct.

v11 p.27 / course p.28. Complete quantifiers, uniqueness and occupational clauses: appendix.

Three types make the wage equations tangible

OUR FINITE ECONOMY

Type	z	Mass
Low	0	.55
Middle	$1/2$	.30
High	1	.15

$$h = 1/2, \quad \mu = 2.$$

ZERO PROFIT BEFORE AI

$$u + \frac{1}{2}v = \frac{1}{2}, \quad u + \frac{1}{2}t = 1, \quad v + \frac{1}{4}t = 1.$$

Subtract the first two: $t - v = 1$. Then

$$(u, v, t) = (.2, .6, 1.6).$$

Worker masses ($L \rightarrow M, L \rightarrow H, M \rightarrow H$) are (.32, .23, .14); all resources clear.

The finite economy is derived directly. It changes the source's continuous-density assumption.

Complete resource equations and no-entry checks: `analysis/derivation.md`. Baseline output $Y_0 = .53$.

Autonomous AI pins wages through outside options

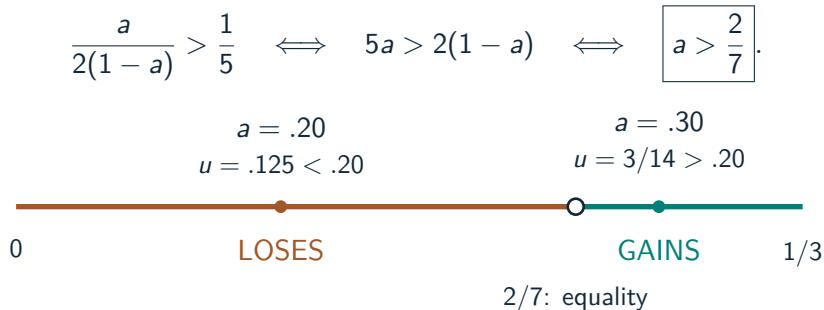
DERIVED BRANCH: $0 < a < 1/3$; **SPARE COMPUTE PINS** $r^A = a$

Active human role	Competitive wage
High: supervise AI workers	$t = (1 - a)/[(1 - a)/2] = 2$
Middle: supervise AI workers	$v = (1 - 2a)/(1 - a)$
Low: work with middle solvers	$u = \frac{1}{2} - \frac{1}{2}v = a/[2(1 - a)]$

.55 low workers use .275 middle solvers. The remaining .025 middle humans supervise AI, pinning v .

AI-worker compute use $7/[20(1 - a)] < 2$; spare compute and all no-entry inequalities checked.

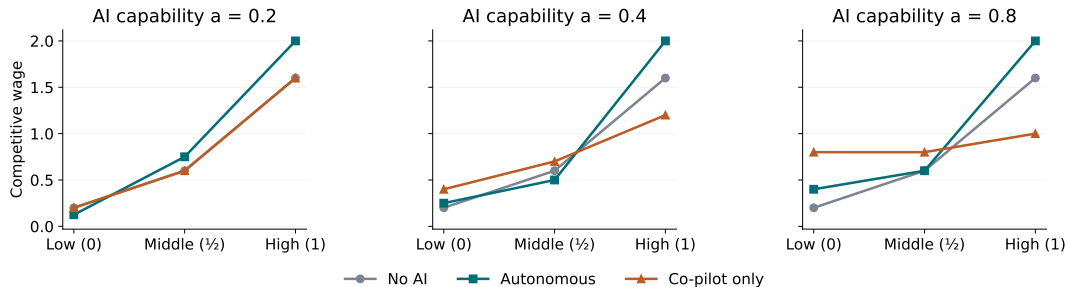
Same autonomy; the bottom changes from loss to gain



The denominator is positive. At $a = 2/7$ there is *no strict gain*. Lean will check exactly this inequality and its domain.

Schematic line; positions are illustrative. $2/7$ is our finite-branch threshold, not the continuum's numerical cutoff.

The numerical check captures both dimensions



Read across panels: change capability while holding autonomy fixed.

Read within a panel: change autonomy at the same capability and compute.

Seven equilibria. Feasibility, no profitable entry, complementary slackness, duality and wage uniqueness checked. Lines join three types only.

More output need not mean more human wage income

HOLD $a = .4$ AND $\mu = 2$ FIXED ACROSS AI REGIMES

	Labor income	Compute income	Total output
No AI	.5300	0	.5300
Co-pilot only	.6100	0	.6100
Autonomous	.5875	.8000	1.3875

WHY OUTPUT RISES

Autonomy allows every co-pilot allocation plus independent AI production. If $a > 0$, idle compute $\delta > 0$ adds $a\delta > 0$.

At $a = 0$, independent output cannot establish strictness; AI workers assisted by humans supply a separate mechanism.

WHY DISTRIBUTION DIFFERS

At $a = .4$, co-pilot wages are $(.4, .7, 1.2)$; autonomous wages are $(.25, .5, 2)$. Expertise and compute ownership matter.

Lean is one link in the verification chain



A HUMAN / SOURCE QUESTION

Does the formal statement capture the paper's objects, assumptions, and conclusion?

A LEAN QUESTION

Does the proof establish that exact formal statement from its stated assumptions?

A green build answers the second question for compiled proofs. It does not automatically answer the first or establish full-paper coverage.

This run: source-pinned algebraic support and a finite example; no completed continuum proposition Specs.

The source equation becomes a precise contract

$$\text{Paper: } n(z) = \frac{1}{h(1-z)}, \quad n(z)(a-w) - r = 0 \implies w = a - h(1-z)r.$$

Lean contract + key steps of zeroProfitWage

```
def ZeroProfitWageSpec : Prop :=
  ∀ h z a w r : ℝ, 0 < h → h < 1 → 0 ≤ z → z < 1 →
    topAutomatedProfit h z a w r = 0 → w = a - h * (1 - z) * r

have hhne : h ≠ 0 := ne_of_gt hh
have hzne : 1 - z ≠ 0 := ne_of_gt (sub_pos.mpr hz)
unfold topAutomatedProfit span at hprofit
field_simp [hhne, hzne] at hprofit
nlinarith
```

Interpretation: real scalars, explicit bounds, zero profit. Nonzero denominators justify the algebra; Lean checks its conclusion.

v11 pp.10/12. FirmAlgebra.zeroProfitWage: build PASS. This is a conditional wage identity, not an equilibrium theorem.

Read the proof as four mathematical moves

THE RELEVANT PROOF FRAGMENT

```
intro h z a w r hh _ _ hz hprofit
have hhne : h ≠ 0 := ne_of_gt hh
have hzne : 1 - z ≠ 0 := ne_of_gt (
  sub_pos.mpr hz)
unfold topAutomatedProfit span at hprofit
field_simp [hhne, hzne] at hprofit
nlinarith
```

The underscores discard hypotheses unused by this algebra: $h < 1$ and $z \geq 0$. They remain in the contract.

Nonzero denominators are part of the proof, not a silent simplification.

The tactic produces a proof term checked by Lean. Familiar theorem names alone do not establish source fidelity.

WHAT EACH STEP DOES

- 1. Introduce** fix the scalars and premises.
- 2. Prove** $h > 0$ and $z < 1$ imply $h \neq 0$ and $1 - z \neq 0$.
- 3. Expand and clear** replace definitions, multiply out denominators.
- 4. Conclude** polynomial arithmetic gives the wage identity.

The same identity exposes the cost of autonomy

THE CHECKED IDENTITY: $w = a - h(1 - z)r$

AUTONOMOUS RENT: $r = a$

CO-PILOT RENT: $r = 0$

$$w^A = a[1 - h(1 - z)].$$

`autonomous_wage` specializes the identity, then `ring` factors the expression.

$$w^N = a.$$

`nonautonomous_wage` substitutes zero rent; `simp` simplifies.

$$a - a[1 - h(1 - z)] = ah(1 - z) > 0 \quad \text{if } a > 0, h > 0, z < 1.$$

This compares one firm's wage formulas. Lean has **not** shown that these roles are active or that equilibrium rents take those values.

FirmAlgebra.lean: `autonomous_wage`, `nonautonomous_wage`, `opportunity_cost_gap`. All compiled in the original run.

The discrete threshold gets an “if and only if” proof

$$0 < a < \frac{1}{3} \implies \left(\frac{1}{5} < \frac{a}{2(1-a)} \iff \frac{2}{7} < a \right).$$

```
def BottomGainSpec : Prop :=
  ∀ a : ℝ, 0 < a → a < 1 / 3 →
    (1 / 5 < bottomWage a ↔ 2 / 7 < a)

theorem bottomGain : BottomGainSpec := by
  intro a ha hupper
  have hden : 0 < 2 * (1 - a) := by linarith
  unfold bottomWage
  rw [lt_div_iff0 hden]
  constructor <=> intro h <=> linarith
```

`hden` licenses inequality-preserving division. `constructor` splits the equivalence into two implications; `linarith` proves each.

Discrete.bottomGain: build PASS. This checks the candidate-wage threshold; the equilibrium interpretation is a separate obligation.

A missing assumption changes the mathematical claim

WHAT FAILED DURING THE RUN

The first denominator-clearing attempt supplied only a product's nonzero proof. A factor denominator remained.

The successful proof explicitly supplied both $h \neq 0$ and $1 - z \neq 0$.

The new boundary check compiles. It is a counterexample to **unrestricted algebra**, not an admissible economic team. Domain restrictions carry content.

WHAT FAILS IF WE DROP $z < 1$

In Lean, real division is total. At

$$h = \frac{1}{2}, \quad z = 1, \quad a = \frac{1}{2}, \quad w = 1, \quad r = 0,$$

$n(z) = 0$ and profit is zero, but the proposed conclusion would say $1 = 1/2$.

Run history: lean/docs/RUN_REPORT.md. New presentation-only check: analysis/BoundaryCheck.lean; original lean/ snapshot unchanged.

What passed, what failed, and what remains open

BUILD / PROOF EVIDENCE

PASS support modules and full root build.

PASS required paper -fast check, exit 0.

No sorry $\mathbf{Ax}$ in the four inspected endpoints;
only standard Lean foundations.

SOURCE / COVERAGE EVIDENCE

STOPPED semantic preflight, exit 2:
source statement map missing.

0 of 6 continuum propositions proved.

Open measurable matching, equilibrium,
sorting, uniqueness, and source-to-Lean
audits.

The algebra generalizes a, w, r to arbitrary reals. The finite model changes the population domain. Neither result silently inherits the full source theorem.

Actual logs: [lean/docs/checks/](#). A successful fast check can compile an empty source interface; it is not semantic acceptance.

The step to check on a handwritten page

THE CLAIM UNDER SCRUTINY

“Distribution depends on autonomy, not capability.”

The calculation gives a different verdict:

$$\frac{a}{2(1-a)} > \frac{1}{5} \iff a > \frac{2}{7}, \quad 0 < a < \frac{1}{3}.$$

Photo pending. Write and verify the derivation yourself, then save the genuine photograph as `hand/derivation.jpg`. Recompile to show it here.

The typeset exercise supports preparation; it does not substitute for the required handwritten evidence or your own verdict.

The result is stronger when its boundary is explicit

ECONOMIC CONCLUSION

Bottom: capability crosses a threshold.

Top: autonomy scales expertise under $h < h_0$ and $a < 1$.

Output: more feasible uses of compute increase production.

Autonomy and capability both matter. A valid proof is valuable precisely because we can say what it proves—and what remains outside it.

VERIFICATION CONCLUSION

By hand: a finite equilibrium branch.

Numerically: allocations and prices agree.

In Lean: specified algebra follows from explicit assumptions.

Source: Ide & Talamàs (2025), arXiv v11. Repository: github.com/gsaco/ai-05-ide.

Proposition 6: the complete quantifiers

The non-autonomous equilibrium is unique, efficient, maximizes labor income within its technology, and $r^N = 0$. If $a \leq w(0)$, AI is unused. If $a > w(0)$:

$$W_a^N \preceq (W_p^N \cup I^N \cup S_p^N), \quad W_a^N, W_p^N, S_p^N \neq \emptyset;$$

only I^N may be empty. In either case,

$$Y^A > Y^N;$$

$$\exists z \in (0, 1] : w^N(z) \leq w(z) \quad (\text{strict if } a > w(0));$$

$$\begin{aligned} \exists \epsilon > 0 \forall z \in [0, \epsilon) : w^N(z) \geq \max\{w(z), w^A(z)\} \\ (\text{strict if } a > w(0)); \end{aligned}$$

$$\exists \epsilon > 0 \forall z \in (1 - \epsilon, 1] : w^N(z) \leq w^A(z) \quad (\text{strict if } z \neq 1).$$

v11 p.27 / course p.28. W_a : AI-assisted; W_p : human-assisted; I : independent; S_p : human solvers.

What the optimization program actually checks

PRIMAL: CHOOSE ACTIVITIES

$$\max_{x \geq 0} q^\top x \quad \text{s.t. } Ax \leq b.$$

An activity uses human time and compute.
The three human masses and μ form b .

DUAL: PRICE RESOURCES

$$\min_{p \geq 0} b^\top p \quad \text{s.t. } A^\top p \geq q.$$

Payments cover output in every potential firm: no profitable entry. Active firms have equality.

Matching primal output to dual factor income certifies the selected finite equilibrium. Checking the optimal dual wage range avoids mistaking one solver's prices for unique wages.

analysis/discrete_model.py; analysis/results.json; checks/computation.txt. This is not a continuum proof.

Source pin, abundance, and reproducibility

$$\mu > \int_0^a h(1-z) dG(z) + \frac{1-G(a)}{h(1-a)}$$

This is a sufficient abundance condition in the source.

	Course reading	Lean source pin
Date	May 20, 2025	February 25, 2025
Version / length	v12 / 39 pages	v11 / 35 pages
Propositions 5 / 6	pp.25 / 28	pp.24 / 27

Reproduce: make analysis slides. Check Lean in the pinned AppliedModelingLib clone, not as a standalone weekly-folder project.

arxiv.org/abs/2312.05481v11. Submission stamp: February 24 UTC. Citation year 2025; Talamàs uses a grave accent.